Patient Similarity Networks (PSN)

BMED365: Topic List P01–P08

BMED365

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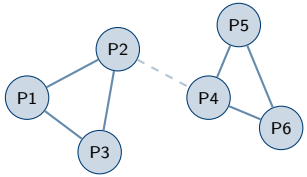
Overview

1. **Fundamentals of PSN**
   - P01: Explain the concept of patient similarity networks (PSN)
   - P02: Describe how PSN can support precision medicine
2. **Similarity Calculation**
   - P03: Calculate similarity between patients
   - P04: Construct a PSN from a patient-feature matrix
   - P05: Choose threshold value for edge creation in PSN
3. **Analysis of PSN**
   - P06: Identify patient subgroups via community detection
4. **Advanced Topics**
   - P07: Advantages and limitations of the PSN approach
   - P08: Multimodal PSN (integration of different data types)

P01: Explain the concept of patient similarity networks (PSN)

What is a patient similarity network?

- A **graph** where each node represents a patient
- **Edges** connect patients who are similar to each other
- Similarity based on clinical variables (symptoms, vitals, demographics), biomarkers, genetics, imaging, lifestyle factors



Two patient groups with strong internal similarity

Main idea:

- Patients with similar profiles are grouped together
- Reveals natural subgroups in the patient population
- Foundation for **patient stratification**: identify subgroups with shared characteristics for targeted treatment

Key references: Pai & Bader (2018), Urda-Garcia et al. (2025), Bhalla et al. (2021)

P02: Describe how PSN can support precision medicine

Precision medicine

Tailoring treatment to the individual patient based on their unique profile – not "one size fits all"

PSN supports precision medicine by:

1. **Identifying subgroups** – Finding patients with similar disease mechanisms
2. **Predicting treatment response** – "Patients who are similar to you responded well to X"
3. **Discovering unknown relationships** – Revealing patterns across data types and established diagnoses
4. **Integrating heterogeneous data** – Combining clinical, omics, imaging data

Example: IBS study

In Lab 1 we use PSN on IBS patients to investigate relationships between gastrointestinal symptoms, brain imaging (MRI), and cognitive functions.

P03: Calculate similarity between patients

Common similarity/distance measures:

Euclidean distance (ℓ_2):

$$d_E = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}$$

- Straight line between points in $\mathbb{R}^n$
- Sensitive to scale
- Good for continuous data

Manhattan distance (ℓ_1):

$$d_M = \sum_{i=1}^{n}|x_i - y_i|$$

- "City block" distance
- More robust to outliers

Gower distance:

$$d_G = \frac{1}{n}\sum_{i=1}^{n} d_i(x_i, y_i)$$

d_i = *normalized distance for variable* i:
num.: range-norm., cat.: 0/1, binary: id.

- Handles **mixed data types**
- Normalizes automatically
- Ideal for clinical data!

Conversion

similarity = 1 − distance (norm. to [0,1])

P04: Construct a PSN from a patient-feature matrix

Step-by-step process:

1. **Organize data:** Patient $\times$ Feature matrix

Patient	Age	Gender	BMI	Blood Pressure	Symptom Score	Biomarker
P1	45	F	24.5	120/80	7	1.2
P2	42	F	25.1	118/78	8	1.1
P3	67	M	31.2	145/95	3	2.8

2. **Normalize/standardize:** Make variables comparable
   (z-score: $(x - \mu)/\sigma$, min-max: $(x - \min)/(\max - \min)$)
3. **Calculate similarity matrix:** All pairs of patients
4. **Choose threshold value:** Determine when two patients are "similar enough"
5. **Create edges:** Connect patients with similarity $>$ threshold
6. **Build graph:** Use NetworkX to create the network

Python code (conceptual)

```
G = nx.Graph()
for i, j:  if similarity[i,j] > threshold:  G.add_edge(i,j)
```

P05: Choose threshold value for edge creation in PSN

Definition

Threshold τ: Edge created between patient i and j if similarity $s_{ij} > \tau$.

The challenge (threshold selection is **critical** for results and interpretation):

- Too low threshold $\rightarrow$ Too many edges, all patients connected
- Too high threshold $\rightarrow$ Too few edges, isolated nodes

Strategies for threshold selection:

1. **Percentile-based:** Keep top 5–10% of edges
2. **k-nearest neighbors (k-NN):** Each node connects to k nearest
3. **Statistical threshold:** $\mu + 1.5\sigma$ of similarity distribution
4. **Visual inspection:** Try different values and visualize

Too many edges

Lose subgroup structure, difficult to interpret

Too few edges

Lose information, fragmented network

P06: Identify patient subgroups via community detection

Community detection in PSN:

- Finds natural groupings of patients
- Patients in the same group are more similar to each other than to others

The Louvain algorithm (from N09):

1. Optimizes **modularity** – internal density vs. external connections
2. Hierarchical: Finds groups at multiple levels
3. Fast and scalable

Clinical interpretation:

- Each community = potential patient subtype
- Examine characteristics for each group
- Compare outcomes and treatment response

Python code

```
# NetworkX (built-in):
nx.community.louvain_communities(G)

# Alternative (python-louvain):
import community
community.best_partition(G)
```

P07: Advantages and limitations of the PSN approach

Advantages:

- ✔ Intuitive and visual
- ✔ Integration of different data types
- ✔ No assumption about underlying distributions
- ✔ Discovers non-linear relationships
- ✔ Network analysis tools can be applied
- ✔ Patient-centered perspective

Limitations:

- ✖ Choice of similarity measure and threshold is critical
- ✖ Scales poorly to large datasets
- ✖ Missing data can be problematic
- ✖ Interpretation challenges
- ✖ Not always reproducible
- ✖ Requires domain expertise

Summary

PSN is a powerful tool for exploratory analysis and hypothesis generation, but results should be validated with independent data and methods.

P08: Multimodal PSN (integration/fusion of different data types)

Multimodal PSN:

- Combines data from **multiple sources/modalities**
- Examples: Clinical + Genetics + Imaging

Integration strategies:

1. **Early fusion:** Combine all features before similarity calculation
2. **Late fusion:** Create separate PSNs, combine afterwards
3. **Intermediate fusion:** Weighted combination of similarity matrices

Example from Lab 1 (IBS study):

- **Clinical data:** Symptom scores, demographics
- **MRI data:** Brain imaging (connectivity, volume)
- **Cognitive data:** Test results
- $\rightarrow$ Integrated PSN for holistic patient profiling

Value of multimodal PSN

Captures the complexity in patient populations that is not visible in single modalities alone.

Summary: PSN

Key points:

- **P01:** PSN = Graph where nodes are patients, edges are similarity
- **P02:** Supports precision medicine via subgroup identification
- **P03:** Similarity measures: Euclidean, Manhattan, Gower
- **P04:** Construction: Feature matrix $\rightarrow$ Similarity $\rightarrow$ Thresholding $\rightarrow$ Graph
- **P05:** Threshold selection is critical for network structure
- **P06:** Community detection (Louvain) for subgroups
- **P07:** Advantages vs. limitations must be weighed
- **P08:** Multimodal PSN integrates heterogeneous data

Lab 1 – Practical experience

Construct and analyze PSN with real clinical data (IRIS, IBS, WAIS-IV) using Python and NetworkX.